## ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 2 | Alternate digit sums must differ by 0 or 11 |
| 2 | **(a)** 0 | The number must be even and its digit sum a multiple of 3 |
| 3 | **(c)** 8262 | Largest arrangement minus smallest arrangement |
| 4 | **(c)** 60 | Find the LCM |
| 5 | **(b)** 3 | 4829 leaves remainder 5 when divided by 8 |
| 6 | **(a)** 1001 | Smallest 4-digit number is 1000; add what is needed to reach a multiple of 7 |
| 7 | **(a)** 45 | Subtract the reverse |
| 8 | **(c)** 2 | Alternate digit sums must differ by 0 or 11 |
| 9 | **(c)** 0 | The number must be even and its digit sum a multiple of 3 |
| 10 | **(a)** 6084 | Largest arrangement minus smallest arrangement |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | 8456 divided by 11 leaves remainder 8 (1); add $11 - 8 = 3$ (1). | 1+1 |
| 12 | By 3: Yes, digit sum 15 is a multiple of 3 (1); by 4: Yes, last two digits 16 are divisible by 4 (1). | 1+1 |
| 13 | 842-248=594 (1); $594 = 99 \times 6$, so divisible (1). | 1+1 |
| 14 | 3721 divided by 8 leaves remainder 1 (1); add $8 - 1 = 7$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | LCM of 12 and 15 is 60 (1); first multiple not below 1000 (1); 1020 (1). | 1+1+1 |
| 16 | Alternate sums for each trial; need difference 0 or 11 (1); * $= 0$ (1); check 8305 (1). | 1+1+1 |
| 17 | Middle number $= \frac{54}{3} = 18$ (1); numbers 17, 18, 19 (1); check sum $= 54$ (1). | 1+1+1 |
| | **OR:** LCM of 14 and 21 is 42 (1); first multiple not below 1000 (1); 1008 (1). | 1+1+1 |
| 18 | Alternate sums for each trial; need difference 0 or 11 (1); * $= 9$ (1); check 6919 (1). | 1+1+1 |
| | **OR:** Middle number $= \frac{81}{3} = 27$ (1); numbers 26, 27, 28 (1); check sum $= 81$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Steps: 9541-1459=8082; 8820-0288=8532; 8532-2358=6174 (3); total 3 steps (2). | 3+2 |
| | **OR:** LCM(6, 8, 12) $= 24$ (3); required number $= 24 + 3 = 27$ (2). | 3+2 |
| 20 | Steps: 8620-0268=8352; 8532-2358=6174 (3); total 2 steps (2). | 3+2 |
| | **OR:** LCM(10, 15, 20) $= 60$ (3); required number $= 60 + 4 = 64$ (2). | 3+2 |

**Mistakes to avoid:**

- Testing 11 with the digit *sum*; the test uses the **difference** of the sums at odd and even places.
- Saying a number is divisible by 6 because it is divisible by 2 or 3; it must be divisible by **both**.
- Thinking that divisibility by 4 needs the digit sum; look at the last **two** digits (last **three** for 8).
- Subtracting in the wrong order in Kaprekar's routine: always take (largest arrangement) − (smallest arrangement).
- Ignoring the leading zero in the smallest arrangement (e.g. 0378 is 378), which gives a wrong difference.
- Assuming a number divisible by 4 and 6 must be divisible by 24; check with 12 (divisible by both, not by 24).

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 24 | Find the LCM |
| 2 | **(c)** 8 | 6713 leaves remainder 3 when divided by 11 |
| 3 | **(c)** 1008 | Smallest 4-digit number is 1000; add what is needed to reach a multiple of 12 |
| 4 | **(b)** 63 | Subtract the reverse |
| 5 | **(d)** 4 | Alternate digit sums must differ by 0 or 11 |
| 6 | **(c)** 2 | The number must be even and its digit sum a multiple of 3 |
| 7 | **(c)** 6264 | Largest arrangement minus smallest arrangement |
| 8 | **(d)** 180 | Find the LCM |
| 9 | **(d)** 1 | 5207 leaves remainder 5 when divided by 6 |
| 10 | **(a)** 1005 | Smallest 4-digit number is 1000; add what is needed to reach a multiple of 15 |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | By 3: Yes, digit sum 21 is a multiple of 3 (1); by 4: Yes, last two digits 48 are divisible by 4 (1). | 1+1 |
| 12 | 753-357=396 (1); $396 = 99 \times 4$, so divisible (1). | 1+1 |
| 13 | 6094 divided by 12 leaves remainder 10 (1); add $12 - 10 = 2$ (1). | 1+1 |
| 14 | By 3: Yes, digit sum 18 is a multiple of 3 (1); by 4: No, last two digits 50 are not divisible by 4 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | LCM of 16 and 24 is 48 (1); first multiple not below 1000 (1); 1008 (1). | 1+1+1 |
| 16 | Alternate sums for each trial; need difference 0 or 11 (1); * $= 5$ (1); check 8547 (1). | 1+1+1 |
| 17 | Middle number $= \frac{111}{3} = 37$ (1); numbers 36, 37, 38 (1); check sum $= 111$ (1). | 1+1+1 |
| | **OR:** LCM of 18 and 27 is 54 (1); first multiple not below 1000 (1); 1026 (1). | 1+1+1 |
| 18 | Alternate sums for each trial; need difference 0 or 11 (1); * $= 0$ (1); check 5082 (1). | 1+1+1 |
| | **OR:** Middle number $= \frac{150}{3} = 50$ (1); numbers 49, 50, 51 (1); check sum $= 150$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Steps: 8731-1378=7353; 7533-3357=4176; 7641-1467=6174 (3); total 3 steps (2). | 3+2 |
| | **OR:** LCM(9, 12, 18) $= 36$ (3); required number $= 36 + 5 = 41$ (2). | 3+2 |
| 20 | Steps: 9542-2459=7083; 8730-0378=8352; 8532-2358=6174 (3); total 3 steps (2). | 3+2 |
| | **OR:** LCM(8, 12, 20) $= 120$ (3); required number $= 120 + 7 = 127$ (2). | 3+2 |

**Mistakes to avoid:**

- Testing 11 with the digit *sum*; the test uses the **difference** of the sums at odd and even places.
- Saying a number is divisible by 6 because it is divisible by 2 or 3; it must be divisible by **both**.
- Thinking that divisibility by 4 needs the digit sum; look at the last **two** digits (last **three** for 8).
- Subtracting in the wrong order in Kaprekar's routine: always take (largest arrangement) − (smallest arrangement).
- Ignoring the leading zero in the smallest arrangement (e.g. 0378 is 378), which gives a wrong difference.
- Assuming a number divisible by 4 and 6 must be divisible by 24; check with 12 (divisible by both, not by 24).

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** 27 | Subtract the reverse |
| 2 | **(b)** 3 | Alternate digit sums must differ by 0 or 11 |
| 3 | **(a)** 1 | The number must be even and its digit sum a multiple of 3 |
| 4 | **(c)** 9081 | Largest arrangement minus smallest arrangement |
| 5 | **(b)** 70 | Find the LCM |
| 6 | **(d)** 8 | 9154 leaves remainder 1 when divided by 9 |
| 7 | **(a)** 1008 | Smallest 4-digit number is 1000; add what is needed to reach a multiple of 18 |
| 8 | **(d)** 9 | Subtract the reverse |
| 9 | **(a)** 7 | Alternate digit sums must differ by 0 or 11 |
| 10 | **(d)** 2 | The number must be even and its digit sum a multiple of 3 |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | 961-169=792 (1); $792 = 99 \times 8$, so divisible (1). | 1+1 |
| 12 | 5518 divided by 7 leaves remainder 2 (1); add $7 - 2 = 5$ (1). | 1+1 |
| 13 | By 3: Yes, digit sum 15 is a multiple of 3 (1); by 4: Yes, last two digits 24 are divisible by 4 (1). | 1+1 |
| 14 | 634-436=198 (1); $198 = 99 \times 2$, so divisible (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | LCM of 20 and 30 is 60 (1); first multiple not below 1000 (1); 1020 (1). | 1+1+1 |
| 16 | Alternate sums for each trial; need difference 0 or 11 (1); * $= 8$ (1); check 3784 (1). | 1+1+1 |
| 17 | Middle number $= \frac{213}{3} = 71$ (1); numbers 70, 71, 72 (1); check sum $= 213$ (1). | 1+1+1 |
| | **OR:** LCM of 9 and 25 is 225 (1); first multiple not below 1000 (1); 1125 (1). | 1+1+1 |
| 18 | Alternate sums for each trial; need difference 0 or 11 (1); * $= 6$ (1); check 9636 (1). | 1+1+1 |
| | **OR:** Middle number $= \frac{333}{3} = 111$ (1); numbers 110, 111, 112 (1); check sum $= 333$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Steps: 7610-0167=7443; 7443-3447=3996; 9963-3699=6264; 6642-2466=4176; 7641-1467=6174 (3); total 5 steps (2). | 3+2 |
| | **OR:** LCM(4, 14, 21) $= 84$ (3); required number $= 84 + 2 = 86$ (2). | 3+2 |
| 20 | Steps: 8621-1268=7353; 7533-3357=4176; 7641-1467=6174 (3); total 3 steps (2). | 3+2 |
| | **OR:** LCM(12, 16, 24) $= 48$ (3); required number $= 48 + 6 = 54$ (2). | 3+2 |

**Mistakes to avoid:**

- Testing 11 with the digit *sum*; the test uses the **difference** of the sums at odd and even places.
- Saying a number is divisible by 6 because it is divisible by 2 or 3; it must be divisible by **both**.
- Thinking that divisibility by 4 needs the digit sum; look at the last **two** digits (last **three** for 8).
- Subtracting in the wrong order in Kaprekar's routine: always take (largest arrangement) − (smallest arrangement).
- Ignoring the leading zero in the smallest arrangement (e.g. 0378 is 378), which gives a wrong difference.
- Assuming a number divisible by 4 and 6 must be divisible by 24; check with 12 (divisible by both, not by 24).

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.